MODEL X 2.0 Powered By Zuu Crew

PHASE 1 · PROJECT PROPOSAL

Team name Price Signal

IEEE Computational Intelligence Society Student Chapter, IIT

Choose one domain

☑	**01**	Cost of living and everyday economics
☐	**02**	Climate, environment and disaster risk
☐	**03**	Health and wellbeing - non-clinical only: no diagnosis, no treatment advice

1. Problem statement and definition

Small canteens, restaurants and shops in Colombo buy vegetables several times a week while prices can change sharply. A buyer can see today's supplier or market quote, but cannot readily tell whether a movement at Dambulla - the country's major wholesale hub - is likely to reach Pettah or Narahenpita over the next few report days. Buying before a fall or after a rise can directly reduce an already thin operating margin.

Who is affected, and how many. Small canteens, restaurants and shops in Colombo that buy vegetables several times a week. The project does not claim a population count because no reliable count was found. Significance instead rests on the frequency of the decision, perishability, limited cash flow and the immediate effect of price changes on small businesses.

What it costs them today. Buying a day too late after a rise, or too early before a fall, costs money directly. For illustration only: Pettah retail lime went from 1,000 to 1,100 Rs/kg on 9 Oct 2026, so a buyer who needed 10 kg and bought that day instead of the day before paid Rs 1,000 more. Across our 2024 to 2026 test data, a buyer with perfect hindsight about the next 3 report days would have paid 3.6 to 5.4% less on average than one who always buys at once. That is the upper limit of what better timing can save.

How they deal with it now, and why that is not enough. Buyers use supplier quotes, current market prices, other traders, recent experience, stock and expected demand. What is missing is a tested view of whether Dambulla adds advance information, how large the possible move is and when not to trust it.

Problem boundary

- The system supports timing and checking; it never commands the user to buy, wait or sell.
- It covers only foods with a usable Dambulla wholesale series and a supported Colombo price series.
- It estimates movement over report days, not calendar days, and a three-report-day horizon may cross a weekend or holiday.
- It does not claim observed business savings; the financial evaluation is a simulation with explicit assumptions.

2. The user

The named user is a Colombo small-food buyer responsible for frequent vegetable purchases without an analyst or procurement system.

Question	Answer
What decision?	Buy the full requirement today, split the purchase, or wait and re-check.
How often?	Daily or several times per week, depending on the business.
What is used now?	Supplier quotes, current market prices, other traders, recent experience, stock and expected demand.
What is missing?	A tested view of whether Dambulla adds advance information, how large the possible move is and when not to trust it.
What changes?	The buyer gains an additional evidence source, while still confirming the actual supplier price before purchasing.

What the user sees if the system exists. For each item, one card: whether their market's price is lagging behind Dambulla, how much of that gap has typically closed within 3 report days, how uncertain that is, and how old the data is. When the data cannot support a view, the card says "no clear signal". The system supports the decision. It does not tell anyone what to buy.

3. Literature review

Research confirms that Dambulla and regional vegetable prices are linked, and several tools publish the prices, but we found nothing that turns the link into a daily, tested warning for a Colombo buyer.

Published work

- **Spatial integration of vegetable markets during COVID-19 [2].** Uses Engle-Granger cointegration to compare nine regional wholesale markets with Dambulla. During the pandemic only beans in a few markets, Colombo among them, moved together with Dambulla. Before it, most regional markets for beans, carrot, tomato and brinjal were integrated with Dambulla in the long or short run, except Ampara, Colombo and Dehiaththakandiya. It falls short for our user because it tests whether markets are linked, in wholesale prices, and does not forecast, cover retail prices, or serve a buyer.
- **Spatial linkages of green chili prices [3].** Compares Dambulla with 25 regional markets over 2012 to 2016 using Johansen cointegration. All pairs were linked in the long run, Dambulla played a central role in price discovery, and adjustment back to equilibrium after a short-term gap was slow. It covers one commodity and is a historical analysis, with no out-of-sample warning.
- **Price fluctuations of green chilli and tomato at Dambulla [4].** Fits SARIMA and ARIMA forecasts to ten years of monthly Dambulla wholesale prices and tests causality between the two vegetables. It forecasts Dambulla from its own history at monthly frequency, and does not use Dambulla to anticipate prices in Colombo.

Existing tools and products

- **Central Bank Daily Price Report [1].** Authoritative wholesale and retail prices for selected items in several markets, published as a daily PDF. It compares each price only with the previous report, gives no forecast or uncertainty, and the format is hard to use in a program.

- **HARTI price publications [5].** The institute has collected food prices since 1980 and publishes weekly and monthly wholesale and retail prices for Colombo and suburbs. The cadence is weekly or monthly and the output is tables of past prices. We have not confirmed which of these bulletins are still published today.
- **FAO Food Price Monitoring and Analysis tool [6].** Lets users scan prices, see percentage changes, compare markets and count price anomalies in a series. It is built for countries and organisations, and an anomaly count within one series does not say whether a source market is ahead of a destination market. We have not confirmed its coverage of Sri Lankan markets.
- **Prior competition entries.** We could not see earlier ModelX entries, so we make no claim about them.

The gap we step into. Published work establishes the long-run link, and tools publish the prices. In what we found, nobody (1) uses the daily gap between a Colombo price and its Dambulla price to estimate how much Colombo will close in the next few days, (2) tests that on unseen data against a strong baseline with honest uncertainty, and (3) packages it for a buyer with rules for saying nothing. Our early results (section 4) agree with the slow adjustment reported in [3]. Study [2] found Colombo less tightly linked to Dambulla than most markets before the pandemic. Our data start in July 2020, after the pandemic began, so we cannot test the earlier period, and a change in market conditions could weaken the signal (section 10).

4. Proposed solution

What the system does. Each working day it reads the latest Central Bank price report and compares every Colombo price (wholesale Pettah, retail Pettah, retail Narahenpita) with the Dambulla wholesale price for the same item. For each of about 18 vegetables and fruits, it tells a Colombo buyer whether Colombo is lagging behind Dambulla and how much of that gap has typically closed within the next 1, 3 and 5 report days. If the data are stale or too thin, it says it cannot tell. The data come from the Statistics Department, Central Bank of Sri Lanka, which does not endorse this project or its findings (section 5.1).

The insight it produces. A pattern: when a Colombo price sits unusually far from the Dambulla price, Colombo tends to close part of the gap within days, so Dambulla gives advance information about Colombo. Early evidence comes from a test on data the model never saw (training to 2023, testing 2024 to October 2026).

Adding the Dambulla price improved forecasts of Colombo price changes beyond a model that already uses the price's own history, including its pull back toward its own average.

Colombo series	Horizon (report days)	Gain (points)	95% interval	Placebo	Items with gain	Verdict
Wholesale Pettah	1	+7.3	+6.4 to +8.1	+0.0	14 of 18	supported
Wholesale Pettah	3	+9.5	+7.7 to +11.3	+0.1	15 of 18	supported
Wholesale Pettah	5	+9.0	+6.8 to +11.3	+0.2	15 of 18	supported
Retail Pettah	1	+6.4	+5.5 to +7.3	+0.0	14 of 19	supported
Retail Pettah	3	+8.4	+6.5 to +10.2	+0.0	16 of 19	supported
Retail Pettah	5	+7.9	+5.6 to +10.2	+0.1	16 of 19	supported
Retail Narahenpita	1	+7.5	+5.9 to +8.9	-0.1	10 of 19	partly
Retail Narahenpita	3	+12.2	+10.5 to +14.0	-0.3	16 of 19	supported
Retail Narahenpita	5	+12.0	+9.8 to +14.1	-0.2	17 of 19	supported

Gain is the share of squared forecast error removed, in percentage points, compared with a model that uses only the price's own history. The placebo swaps in a different item's Dambulla price and should show nothing. "Supported" means the interval is above zero, the placebo gain is under half the real gain, and at least 60% of items improve. The gain was similar in each test year for every row except one: wholesale Pettah at 3 days was +9.5 in 2024, +9.6 in 2025 and +9.1 in 2026, but retail Narahenpita at 1 day fell from +8.2 and +9.0 to +3.3 in 2026, which is why it is marked "partly". In the fitted models, roughly a fifth to a quarter of an unusual gap closed within 3 report days (gap coefficients of 0.17 to 0.25). These are early results from one train and test split, and section 9 lists what they do not show.

What it is worth in rupees. We simulated a buyer who must buy an item once, either today or 3 report days later, and who follows the model's call: wait only when it predicts a fall beyond a threshold fixed on pre-2024 data, otherwise buy today. Over the same test period, following the model that uses Dambulla was cheaper than always buying today by 1.2 to 1.6% of the purchase value, about Rs 60 to 96 per 10 kg. It was also cheaper than following a model that uses only the price's own history, by 1.0 to 1.6 points, and every 95% interval was below zero. Acting only on the strongest 25% of calls, the direction was right 75 to 85% of the time, against 62 to 69% over all calls. Perfect hindsight would have saved 3.6 to 5.4%, so the signal captures roughly 27 to 44% of what better timing could offer.

Colombo series	Dambulla model vs always buying today	Own-history model vs always buying today	Hindsight ceiling	Dambulla vs own history (95% interval)	Rs per 10 kg, Dambulla model
Wholesale Pettah	-1.43%	+0.14%	-5.35%	-1.57 pts (-1.98 to -1.18)	-62
Retail Pettah	-1.23%	-0.06%	-4.41%	-1.16 pts (-1.47 to -0.87)	-60
Retail Narahenpita	-1.57%	-0.52%	-3.57%	-1.05 pts (-1.24 to -0.85)	-96

Negative means cheaper. This is a simulation, not observed savings. It assumes the buyer can defer a purchase by up to 3 report days at no extra cost, which small kitchens and perishable produce may not allow, and it ignores transport and time. Ten kilograms per decision is an illustrative size. Consecutive decisions overlap, so the month-resampled intervals are the honest guide, not the point estimates.

Why the user could not reach this alone. To see it, a buyer would need years of daily prices for every item in two markets, the gap measured against each item's own usual margin, and a test on unseen data to separate a real lead from noise and from the price's own tendency to return to its average. A buyer sees one day's quote, and the Central Bank PDF shows one day's change.

How it fails safely. Every output shows the date of the latest report used. If that report is older than 3 working days, or the parser rejects the newest report, the system declines to estimate and says why. Items with too little recent data, or a closed market that day, show "insufficient data". When the gap is too small for the model to have shown skill, the card says "no clear signal". The 80% intervals are calibrated on held-out data and, in our first test, contained 77.4 to 81.5% of the outcomes the model had not seen; they will be widened or withdrawn if coverage is poor. The interface states that results are statistical estimates, not advice.

5. Data

5.1 Source and access

What it is and where it comes from. The Statistics Department, Central Bank of Sri Lanka, Daily Price Report [1], published at cbsl.gov.lk/en/statistics/economic-indicators/price-report. Each report in the current layout is a two-page PDF: page 1 summarises the largest moves, and page 2 is a table of wholesale and retail prices for about 40 items (vegetables, other staples, fruit, rice and fish) in Pettah, Dambulla, Narahenpita and the fish markets of Peliyagoda and Negombo, with the previous reported price beside each.

Evidence that we can obtain it. The index page lists about 2,270 daily reports from 2017 to mid-October 2026, 12 per page, each linked directly to its PDF. Our downloader has retrieved them, and the repository's manifest records the URL, date, file hash and download time of each. Our daily refresh script has since added newer reports, the latest dated 15 Oct 2026. Sample rows from our extracted table for 9 Oct 2026:

Item	Price type	Market	Previous (Rs)	Today (Rs)
Lime	retail	Pettah	1,000	1,100
Lime	retail	Dambulla	1,080	1,255
Salaya	wholesale	Negombo	320	550

In the analysis run on data to 13 Oct 2026, the extractor produced 268,577 price rows from the reports it reads. Its output matches 125 figures quoted in the reports' own summary text (automated spot checks in the repository), and each report's prior prices are checked against the previous report's prices (about 0.5% of compared pairs differ, and we flag them). Coverage at that cut: 1,487 of 1,495 in-scope reports parsed (28 July 2020 to 13 October 2026; coverage.csv and the README in the repository). The 8 not parsed are 4 scanned PDFs with no text layer, 3 whose header names a weekday but no date, and 1 file that duplicates the previous day's report. All are listed in the repository, and none is guessed.

How much, and what it covers. 1,487 working-day reports from 28 July 2020 to 13 October 2026 are in scope, which is the period whose layout the parser reads. 776 older reports (2017 to mid-2020) use other layouts and are not used.

Origin and personal data. The Statistics Department publishes the prices; we have not confirmed how it collects them. The data are prices by item and market only. They contain nothing about identifiable people.

Right to use. The Daily Price Report is published openly on the Central Bank's website. On 11 Oct 2026 we wrote to the Statistics Department to ask whether this use was acceptable. The Director replied on 12 Oct 2026 that the data is publicly available and does not require specific approval or permission for use [7]. The Department asked us to cite the source as "Statistics Department, Central Bank of Sri Lanka", and stated that the Central Bank bears no responsibility for, and does not endorse, the content, findings or interpretations of any project derived from the reports. This project is therefore not endorsed by the Central Bank, and we say so wherever we present results. We also follow robots.txt (10-second crawl delay), keep the full set of raw PDF files out of the repository (only a small sample used by the tests is kept there, with attribution), and use the data only for this non-commercial, educational project.

5.2 Extraction method

How. Scraping and PDF parsing. A downloader reads the index page, obeys robots.txt including its crawl delay, downloads each PDF once, keeps the raw files untouched, and writes a manifest. A parser then reads each price table by the horizontal position of every value, because the PDF text layer drops empty cells and splits numbers, which would put prices under the wrong market. Market names come from each section's own header row, and any layout the parser was not built for is rejected, never guessed.

Why this and not the alternatives. We did not find an API or structured download for these prices. Manual collection of about 1,500 reports is not feasible. We could ask the Statistics Department whether a structured file exists, but we cannot depend on a reply, so we parse the published PDFs.

Cleaning, and what we expect to be wrong.

- Item names drift across years (for example "Dhal" and "Red Dhal", "Katta" and "Katta (Imp)"), so we normalise them and log every mapping.
- "n.a." (CBSL reported no price) is kept distinct from a blank cell (not applicable) and from a closed market, which the report's footnote records.
- Missing days (weekends, holidays) are logged and never filled in.
- Prices are rounded to the nearest Rs 5 or 10, so cheap items move in steps.
- The sources contain errors: one report (2 June 2022) quotes a figure in its summary that disagrees with its own table. We follow the table, which agrees with neighbouring days.
- Monday reports compare with last Friday, and some 2021 reports compare with last Monday. The comparison basis is stored with every row.
- Some column headers are blank in the PDF, so those rows are flagged as assumed and treated as less certain.

5.3 Temporality

Historical, recent and live. Both, for different jobs. Years of history train and test the model, because the Dambulla-to-Colombo pattern can only be learned and checked over many months. The latest report, refreshed each working day, drives what the user sees.

Why that fits the user. A buyer decides today, so a static archive would answer "what happened" and not "what is happening now". A system that used only this week's data could not tell a real lead from noise. The newest report in our table at the time of writing is dated 15 Oct 2026.

Refresh, and when the source is unavailable. A script (refresh.py) checks for a new report each working day and records the result; we plan to run it on a daily schedule with GitHub Actions. If the site is down or the newest PDF cannot be parsed, every card keeps showing the date of the last good report. After 3 working days without a good report the system refuses to estimate and says why.

6. Approach and methodology

The approach. Treat a Colombo price as a follower of the Dambulla price and forecast its next move from the gap. For each Colombo series (wholesale Pettah, retail Pettah, retail Narahenpita) and each horizon of 1, 3 and 5 report days, we fit a ridge regression on log price changes, pooled across items. The inputs are the series' own recent changes, its distance from its own 20-day mean, the recent changes in the Dambulla price, and the Dambulla-minus-Colombo gap measured against its own trailing 60-day average. Training uses data to 2023, and testing uses 2024 onward.

Why it fits this problem. The gap term is an error-correction term, the same mechanism studied in the cointegration work on Dambulla and regional markets [2][3], so the model has a plain reading for a buyer: part of the gap tends to close. It has few parameters, which suits more than 1,300 daily observations per series with rounded prices, and each number can be explained. Measuring the gap against its own trailing average stops a fixed Colombo-over-Dambulla margin from passing as a signal.

Alternatives we considered and set aside.

- **Per-item SARIMA or ARIMA [4].** These use only an item's own history, so by design they cannot see a lead from Dambulla, and item-by-item fits are noisy.
- **Cointegration and vector error-correction models [2][3].** The right idea, but built to test whether markets are linked, pairwise and in-sample. We keep the error-correction idea and add out-of-sample testing.
- **Predicting whether a one-day jump will last.** We tested this early. About 57% of jumps persisted, and our models did no better than always guessing persistence (accuracy 56.6 to 56.7% against 57.2%, AUC 0.50 to 0.54). We set it aside and kept the result (jump_check.py and model_check.py in the repository).
- **Gradient boosting or neural networks.** We tested gradient boosting as a challenger under a rule fixed in advance (adopt only if it beats ridge at 3 report days in every Colombo series). It did not: it was 0.6 to 1.4 points worse than ridge, so we kept ridge. We have not tested neural networks, which with this little data risk overfitting and are harder to explain.

How we will work, and what we test as we go.

1. Data pipeline and parser, with tests (done).
2. Feasibility checks of the lead-lag idea, with a placebo and a synthetic self-test whose answer we know (done).
3. Re-run the checks on the final data.
4. Add prediction intervals calibrated on a held-out period (first version done: 77.4 to 81.5% of unseen outcomes fell inside the 80% range).
5. Backtest the decision rule in rupees (first version done, results in section 4), then repeat it on the final data.
6. Build and test the stale-data and refusal rules (built and tested; the website withholds every number when data are more than 3 working days old).
7. Build the interface (first version built, section 7.3).

Throughout: splits are always by time and never use future data, labels that reach past the training cut-off are dropped, the extractor is checked against figures quoted in each report's own text, and consecutive reports must agree with each other.

7. System design

7.1 Architecture

The system is a pipeline with a guard in front of the user. Data moves in one direction, and nothing reaches the user without passing the guard.

1. **Collector.** Reads the Central Bank index page, downloads new PDFs at the site's required pace, keeps the raw files untouched and appends to a manifest.
2. **Parser.** Reads each PDF's price table by column position and writes tidy rows. Reports in unknown layouts go to a quarantine list instead of the table.
3. **Price store.** One table of date, item, market and price type, with flags for closed markets, assumed headers and comparison basis. Raw PDFs are not stored in the repository.
4. **Model.** Fitted offline on the history. Each day it computes the gap and the other inputs for every item and produces a catch-up estimate with an interval.
5. **Guard.** Checks that the latest report is fresh, that the parser accepted it, that the item has enough recent data, and that the gap is large enough for the model to have shown skill. It decides whether the user sees an estimate or a reason for none. The web server applies the same checks again on every request.
6. **Interface.** Shows the cards, with the data date on every one.
7. **Scheduler.** Runs steps 1 to 5 each working day and records the outcome (currently a script we run; GitHub Actions is planned).



The guard is the only way from the model to the interface.

7.2 Technical stack

Part	Choice	Why
Language	Python 3	Team already uses it, and every library below is available
Download	requests, BeautifulSoup	Simple, controllable pacing and retries
PDF parsing	pdfplumber	Gives the position of every word, which the parser needs
Data handling	pandas, numpy	Standard for time series and the ridge models
Storage	CSV files for the pipeline; SQLite for the website, built from the model output	No server, easy to inspect and commit
Modelling	Ridge regression (numpy), scikit-learn if we add a challenger	Few parameters, explainable
Interface	A Flask web page backed by SQLite (first version built), with the Streamlit dashboard kept as an alternative	The page loads nothing from other sites and works on a phone; Streamlit was the fast first version
Scheduling	GitHub Actions on a daily schedule (planned)	Free for public repositories
Testing	pytest	Tests exist for the parser, the model, the guard and the web server
Version control	git and GitHub	Public repository with the commit history

Everything above is free and open source. Nothing costs money so far. We have not yet chosen where to host the website for a live demo; the demo can run locally on real data, and we have not confirmed a free host for the Flask version. We have written confirmation from the Central Bank that the data may be used without approval (section 5.1).

7.3 Interface and interaction

How the user reaches it. A single-page website they open on a phone or laptop. At the top it states whether estimates are shown and the date of the latest report. The user picks one of three Colombo prices, filters items by name, and sees all items on one scale, ordered from the largest expected fall to the largest expected rise. Opening an item shows its price history for Dambulla and Colombo, with a table version. Further pages show past prices of any item as a chart (Explore), a weekly comparison with a CSV download and print view (Weekly report), the evidence behind the model (Method) and a help page. All of them obey the same freshness guard.

What they see. One row or card per item. A real example from the 9 Oct 2026 report, retail Pettah:

Field	Potato (Local), retail Pettah
Latest report	9 Oct 2026
Dambulla wholesale	Rs 185
Pettah retail	Rs 367
Gap against usual margin	Pettah is 28.1% below its usual level against Dambulla
Expected Pettah change over 3 report days	-6.5% (80% range -16.0% to +4.1%), strong signal
When the model has nothing to say	“no clear signal” or “insufficient data”, with the reason, and no numbers

The user’s decision is whether to buy the full amount today, split the purchase across days, or wait. The page never says “buy” or “sell”. When the guard refuses, it shows the reason, for example “the latest report is 4 working days old”, in place of any number. It also states that the Central Bank does not endorse the project.

8. Implementation plan

Milestones

Dates	Milestone	Done when
10-19 Oct	Research, data pipeline, parser, feasibility evidence, user discovery and proposal	Proposal submitted; repository public; claims trace to saved evidence
20-26 Oct	Freeze the final data cut, rerun modelling and calibration, close high-risk gaps	Metrics reproduce; 85+ tests and all refusal cases pass
27-29 Oct	Public deployment, wording retest if available, accessibility and signed-out checks	Every page and /healthz work without authentication
30-31 Oct	Technical report, two-page summary and three-minute video from the same build	Files agree on commit, data date, claims and limitations
1 Nov	Implementation submission	Repository, website, report, summary and video open in a private browser
9-14 Nov	If selected, rehearse the live final	Live demo runs on current real data and safe refusal can be explained

What finished looks like (Phase 2 deliverables)

- Public Git repository with a README that runs from a clean machine, pinned dependencies and visible history.
- Public web application using real CBSL data, plus safe local fallback instructions.
- Technical report evaluating the implementation against this proposal, including gaps and negative results.
- Two-page summary and a three-minute video demonstrating the exact submitted build end to end.

Who owns what, and what we cut first. Nivetha owns data, modelling, interface, evaluation, documentation and delivery. If time runs short, cut in this order: challenger models, Sinhala/Tamil versions, automated cloud

refresh, one- and five-day views, then extra buyer sessions. Never cut freshness, refusal rules, data-rights evidence or honest evaluation.

9. Evaluation plan

What "correct" means. For a buyer, correct has four parts. When the system says a Colombo price is likely to move toward Dambulla's, it does so more often and by more than chance and more than simple rules would. The ranges it states contain the real outcome about as often as claimed. When it says nothing, that is because the data cannot support a view. And it never shows a number built from stale or broken data.

What success means, and how we measure it

Dimension	Metric and comparator	Acceptance rule
Forecast insight	Squared-error skill of Model D versus Model C, no-change and placebo; time-ordered test dates	Gain interval above zero in most series, placebo under half the real gain, and at least 60% of items improve
Uncertainty	Observed coverage of the nominal 80% range, by series and year	75% to 85%; investigate or widen outside this band
Decision relevance	Simulated purchase cost versus buy-today, always-wait, own-history and hindsight	Report effect and interval; never translate simulation into guaranteed savings
Extraction	Repository-owned parser tests plus the 125-value original-report validation	All reproducible tests pass; discrepancies are logged, never silently repaired
Safe failure	Twelve deliberately constructed faults	12/12 must refuse or pass exactly as specified
User comprehension	Task completion and 1-5 understanding, trust and usefulness ratings	Report individual scores and difficulties; revise any repeated high-severity misunderstanding

Explicit stop rules

- If the gain interval includes zero for most series, or the placebo exceeds half the real gain, stop claiming a Dambulla lead.
- If interval coverage falls below 70% or above 90%, withdraw the range until recalibrated.
- If any refusal case fails, do not publish the affected output.
- If buyers cannot explain the main card, simplify the product before using it in the video or live pitch.

Fallback product. If the predictive claim fails, the project will ship an honest descriptive version: the latest verified data date, the paired Dambulla-to-Colombo gap and a clear 'no reliable signal' state, with no forecast. This preserves useful data work without presenting an unsupported insight.

Known limits to report regardless of outcome. The study covers 18 to 19 foods, one principal train/test split and a relatively calm 2024-2026 test period. Prices are rounded; report-day horizons cross weekends and holidays; the buyer sample is three; the revised wording has not been retested; and market structure may change during a crisis. The proposal treats these as boundaries, not footnotes to hide.

10. Risks and contingencies

Risk	Control / contingency
Signal weakens or was luck	Rerun time-ordered tests and placebo; use the descriptive fallback if the claim fails.
Market regime changes	Track results by year; refuse incomplete inputs; state that the calm test period may not generalise to crises.
CBSL layout or access changes	Respect current rules, stop on access changes, quarantine unknown layouts and show the last verified date.
Users treat it as advice	No buy/sell instruction, prominent uncertainty, non-endorsement, repeated server-side guard and comprehension testing.
The source becomes unavailable, or robots.txt changes	Stop crawling, keep working from the archive already downloaded, and say plainly in the demo which date the data ends.
Parser coverage falls short, or part of the data proves unusable	Limit the study to the supported period, publish the coverage numbers and the list of unparsed reports, and never fill gaps silently.
A team member drops out	The repository and README are written so the other member can continue, and the cut list in section 8 applies.
Time pressure	Cut challenger models, translations, cloud scheduling and secondary horizons before any safety or evaluation work (cut list in section 8).

11. References

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7. Director, Statistics Department, Central Bank of Sri Lanka. Email reply to N. Ramanan, 12 Oct 2026. Redacted copy: https://github.com/IT25101414/price-signal/blob/main/docs/cbsl_reply_2026-10-12.pdf

Tools and data we use. Python, requests, Beautiful Soup, pdfplumber, pandas, NumPy, scikit-learn, Flask, SQLite, pytest and Streamlit, all open source and listed with versions in the repository's requirements files. Data: Statistics Department, Central Bank of Sri Lanka, Daily Price Reports [1].

Ethics, licensing and responsibility

- No personal data are used. Buyer records are anonymised and published only with permission.
- The original software is MIT-licensed. CBSL data, reports, the Sri Lankan emblem and other third-party material remain under their respective terms.
- Attribution is to the Statistics Department, Central Bank of Sri Lanka. CBSL does not endorse the project or accept responsibility for its derived findings.
- The project author owns every submitted claim and will not describe simulated savings as observed impact.

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